1. Wide Open Slots

The open slots are more commonly used because the coils can be form-wound and pre – insulated before placing in the slots. This type of slots also facilitates easy removal and replacement of defective coils. But this type of slots create uneven distribution of flux, there by producing ripples in the emf wave.

2. Semi-Closed Slots

The semi closed type slots are better in this respect but don't permit the use of forms wound coils, there by complicating the process of winding.

3. Closed Slots

Totally closed slots don't disturb the air gap flux and they are rarely used.

(b) ROTOR

Salient Pole type rotor is used in low and medium speed alternators. It has a large number of projecting poles, having their cores bolted on to a heavy magnetic wheel of cast iron or steel. The diameter of this kind of alternator is large and the length is small. The poles and pole shoes are laminated to reduce the heat due to eddy currents. In large machines, field winding consists of rectangular copper strip wound on edge.

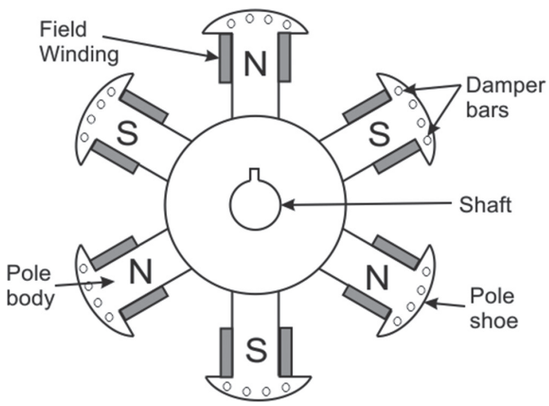


Fig 8.4 *Salient Pole Construction*

8.3.2 Non-Salient Pole Alternator or Turbo Alternator

(i) STATOR

The stator diameter of Non salient pole Alternator is small and the length is long. Stator is the stationary part of the machine, and it is built up of silicon steel laminations having sloth on its inner periphery to house the conductors. Normally this type of alternator has double layer winding. But in high voltage machines, single layer winding is used in semiclosed sloth. Lengthy ventilating ducts are provided for cooling purpose.

(ii) Smooth Cylindrical Type Rotor

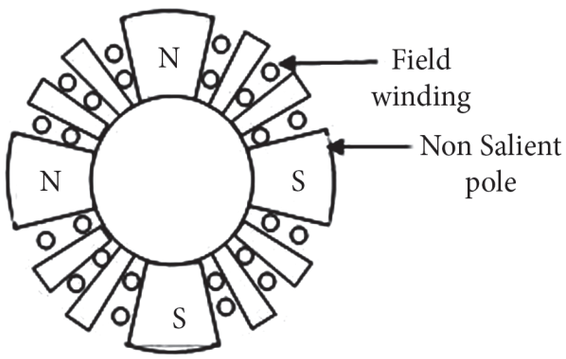


Fig 8.5 *Cylindrical Pole Rotor*

This rotor is used in very high speed alternators driven by steam turbines. The rotor of turbo alternator physically is in the form of smooth cylinder, having long axial length and smaller diameter.

Poles are not projected out from the surface of the rotor. The outer periphery of the rotor stampings has radial slots.

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